

MSC DISSERTATION

Diversification Benefits of Adding Bitcoin to Traditional 60/40 Portfolios

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DECLARATION OF ACADEMIC INTEGRITY

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Abstract

The traditional 60/40 stock-bond portfolio has been the workhorse of institutional and retail asset allocation for over four decades, valued for its predictable risk-return characteristics and its reliance on the historically low correlation between equities and high-grade bonds. The post-2008 macroeconomic environment has progressively challenged this construction: persistent low yields, the 2022 inflation shock that turned stock-bond correlations positive, and successive market dislocations have all prompted institutional investors to search for alternative diversifiers. Bitcoin, once a niche speculative asset, has during this period evolved into a maturing institutional vehicle, particularly following the approval of US spot Bitcoin exchange-traded funds in January 2024. This dissertation investigates whether incremental Bitcoin allocations to a traditional 60/40 portfolio produce statistically significant improvements in risk-adjusted returns, with explicit controls for nonstationarity, regime shifts, and resampling, which are the three methodological concerns identified by the prior academic literature as inadequately addressed.

The empirical analysis uses daily log returns from 2 January 2015 to 31 March 2026 across four series: the S&P 500 index, the Bloomberg US Aggregate Bond Index (LBUSTRUU), Bitcoin (BTC/USD) sourced from CoinDesk, and the three-month US Treasury Bill rate (FRED series DTB3). The sample contains 2,826 trading days aligned on the US equity calendar, with bond and risk-free rate gaps forward-filled. A ten-test empirical battery is structured into five sequential stages: foundation diagnostics, portfolio construction, significance testing, regime analysis, and out-of-sample validation. Foundation diagnostics use the Augmented Dickey-Fuller test to confirm stationarity, with all four return series rejecting the unit-root null at the 1% level, and the Jarque-Bera test to characterise the distributional properties, with all four series exhibiting excess kurtosis above 3.6 and rejecting normality decisively. These results jointly validate the use of Sharpe-ratio-based inference and necessitate non-parametric and non-normal-corrected significance procedures rather than classical parametric tests.

A discrete grid search across Bitcoin allocations from 0% to 10% reveals a monotonically increasing Sharpe-ratio curve, identifying the Sharpe-maximising allocation at the institutional 10% cap with an annualised Sharpe of 0.698 against a benchmark of 0.449, a 55.5% relative uplift. However, because the resampling clause of the research question is given priority over the maximum-uplift figure, the conservative headline allocation adopted is $x^* = 6\%$, the largest weight at which both the block bootstrap and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) cleanly reject the null

hypothesis. At $x^* = 6\%$, the annualised Sharpe rises by 0.18 in absolute terms, a 40% relative uplift over the benchmark, with a one-sided block bootstrap p-value of 0.047 and a Deflated Sharpe Ratio of 0.954, corrected simultaneously for non-normality, sample length, and the multiple-testing bias across ten weight trials. Regime analysis using a 60-day rolling BTC-equity correlation classifier and a Dynamic Conditional Correlation GARCH(1,1) model (Engle, 2002) demonstrates that this benefit is far from uniform: it is concentrated in periods of low correlation, which cover 56% of the sample and in which the Sharpe uplift reaches +0.49, and disappears or reverses during high-correlation regimes. Two pre-specified structural breaks, namely the COVID-19 pandemic of March 2020 and the SEC's spot Bitcoin ETF approval of January 2024, are confirmed by Chow tests at the 1% level.

The walk-forward out-of-sample analysis, conducted across six annual test windows with a rolling five-year training window, qualifies the in-sample result substantially. The optimal allocation identified on training data produces a positive out-of-sample Δ Sharpe in only three of six test windows; the binomial persistence test fails to reject the null at conventional levels, and the stitched out-of-sample 95% bootstrap confidence interval includes zero. The CAPM regression of daily Bitcoin returns on the S&P 500, estimated by ordinary least squares with Newey-West heteroscedasticity-and-autocorrelation-consistent standard errors (Newey and West, 1987), returns a more positive verdict at the asset level: alpha is positive and statistically significant at the 5% level, with an annualised magnitude of 39.3% per year; beta is 0.84, significantly different from zero and borderline below one; and R-squared is 0.052, indicating that approximately 95% of Bitcoin's daily variance is idiosyncratic and unrelated to equity-market movements, which is the necessary condition for genuine diversification value at the asset level.

The dissertation concludes that Bitcoin offers a real but regime-dependent diversification benefit that cannot be reliably harvested by a static allocation rule. The asset itself exhibits the statistical signature of a diversifier; the in-sample portfolio uplift is statistically significant under modern non-normal inference methods; but the absence of out-of-sample persistence at the chosen sample size suggests that a dynamic allocation rule conditioned on the prevailing correlation regime is necessary to capture the benefit reliably. The methodological contribution is to combine the block bootstrap, the Deflated Sharpe Ratio, DCC-GARCH regime analysis, and walk-forward validation within a single integrated framework that addresses each of the three gaps identified in the prior literature.

Keywords:

Keywords: Bitcoin, portfolio diversification, Sharpe ratio, Deflated Sharpe Ratio, regime-dependent allocation

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List of Abbreviations

ADCC	Asymmetric Dynamic Conditional Correlation
ADF	Augmented Dickey-Fuller (unit-root test)
AGG	US Aggregate Bond Index
AIC	Akaike Information Criterion
APA	American Psychological Association
BTC	Bitcoin
CAPM	Capital Asset Pricing Model
CBOE	Chicago Board Options Exchange
CI	Confidence Interval
DCC	Dynamic Conditional Correlation
DSR	Deflated Sharpe Ratio
DTB3	3-Month US Treasury Bill rate (FRED series)
ETF	Exchange-Traded Fund
FRED	Federal Reserve Economic Database
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HAC	Heteroscedasticity and Autocorrelation Consistent (standard errors)
JB	Jarque-Bera (normality test)
LBSTRUU	Bloomberg US Aggregate Bond Total Return Index (ticker)
MSc	Master of Science
NEOMA	NEOMA Business School
OLS	Ordinary Least Squares
OOS	Out-Of-Sample
PBO	Probability of Backtest Overfitting
PSR	Probabilistic Sharpe Ratio
RF	Risk-Free (rate)
RQ	Research Question
SEC	US Securities and Exchange Commission
SPX	S&P 500 index (ticker)
VIF	Variance Inflation Factor
VIX	CBOE Volatility Index

Chapter 1: Introduction

1.1 Background and Context

The 60/40 stock-bond portfolio has been the default asset allocation for institutional and retail investors for several decades. It rests on a simple intuition: equities provide growth and bonds reduce volatility, and because the two have historically moved in different directions, the combination is less risky than either component alone (Markowitz, 1952; Sharpe, 1994). That intuition has weakened since 2008. Persistently low bond yields gave bonds less room to absorb equity drawdowns, the 2022 inflation shock turned the stock-bond correlation positive, and a sequence of market dislocations has forced institutional investors to look for diversifiers that the 60/40 mix no longer reliably provides.

Bitcoin sits awkwardly in this search. It began as a fringe speculative asset and has spent the past decade gradually entering mainstream institutional portfolios, first through ETF analogues and futures products, and most decisively when the US Securities and Exchange Commission approved spot Bitcoin exchange-traded funds in January 2024. Industry research reports that small Bitcoin allocations improved 60/40 Sharpe ratios over the past five years (21Shares, 2025), but whether that improvement is robust enough to justify a permanent place in a traditional portfolio remains contested. That is the question this dissertation attempts to answer.

1.2 Research Problem and Motivation

The academic literature has largely confirmed that small Bitcoin allocations improve in-sample Sharpe ratios. It has been less careful about three related concerns. Most studies report point estimates without significance tests that account for the heavy-tailed return distributions of cryptocurrencies. Most do not condition the result on the prevailing correlation regime, so a benefit concentrated in calm periods is averaged together with periods in which it disappears. And most do not validate their in-sample finding on data that the optimisation never saw. This dissertation addresses all three.

1.3 Research Objectives and Questions

This dissertation pursues the following primary research objective:

The primary research objective is to assess whether adding between 0% and 10% Bitcoin to a traditional 60/40 portfolio improves its risk-adjusted return over the period 2015 to 2026, with explicit controls for nonstationarity, regime shifts, and resampling.

The following research questions guide this investigation:

- RQ1: How does incremental Bitcoin allocation between 0% and 10% affect the annualised Sharpe ratio of a 60/40 portfolio, and at which weight is the improvement maximised?
- RQ2: Is the observed Sharpe-ratio uplift statistically significant under non-parametric resampling and after correction for non-normality and multiple testing?
- RQ3: Does the uplift hold under regime conditioning (using rolling correlation and DCC-GARCH) and under walk-forward out-of-sample validation?

1.4 Scope and Delimitations

The empirical analysis runs from 2 January 2015 to 31 March 2026 on the US equity trading calendar, giving a sample of 2,826 daily observations. The allocation grid is long-only and capped at 10% Bitcoin. Bond exposure is fixed at 40% of the portfolio and Bitcoin substitutes for equity exposure within the equity sleeve. Transaction costs are not modelled, and the analysis does not compare Bitcoin against alternative diversifiers such as gold or commodities; those comparisons are left for future work.

1.5 Significance of the Study

The contribution of this dissertation is more methodological than empirical. The question of whether small Bitcoin allocations improve portfolio Sharpe ratios has been asked before, often with affirmative answers. The novelty here is the combination of four inference procedures applied to the same dataset and the same question: block bootstrap, the Deflated Sharpe Ratio, DCC-GARCH regime analysis, and a walk-forward out-of-sample test. The result, reported transparently in Chapter 4 and discussed in Chapter 5, is more nuanced than the typical institutional headline figure, and is therefore more useful to practitioners who have to defend an allocation decision in front of an investment committee.

1.6 Structure of the Dissertation

The remainder of the dissertation is organised as follows. Chapter 2 reviews the literature and sets out the conceptual framework. Chapter 3 describes the methodology and data. Chapter 4 presents the empirical results across five stages of analysis. Chapter 5 discusses what those results mean in light of the research questions. Chapter 6 concludes with the main findings, the principal limitations, and directions for future research.

Chapter 2: Literature Review

2.1 Theoretical Frameworks

This section sets out the theoretical foundations the empirical chapters rest on. Two strands of theory matter for what follows. The first is modern portfolio theory, which gives the diversification claim its formal basis. The second is the methodological work on the statistical inference of Sharpe ratios and time-varying correlations when returns are neither normally distributed nor independent across time. Both shape the methodological choices made in Chapter 3 and the tests run in Chapter 4.

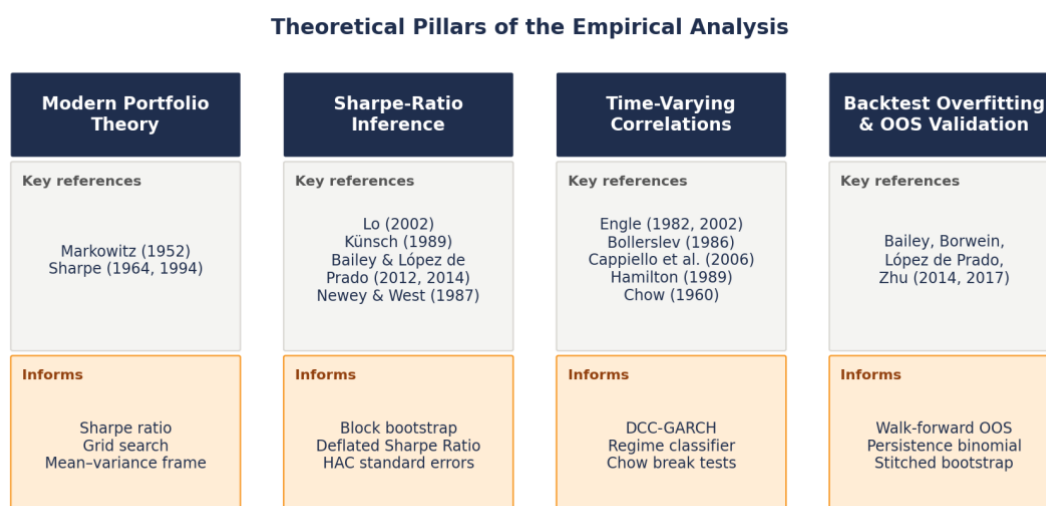
2.1.1 Modern Portfolio Theory and Risk-Adjusted Returns

Modern portfolio theory begins with Markowitz (1952). For any set of risky assets there is a frontier of portfolios that minimise variance for a given expected return, and an asset's contribution to portfolio risk depends not on its own volatility but on its covariance with the rest of the holdings. Assets that move out of step with the existing portfolio reduce overall variance even when they are individually volatile. That is the formal foundation for the diversification claim this dissertation tests. The Sharpe ratio (Sharpe, 1964, 1994) is the standard summary of risk-adjusted return: excess portfolio return divided by total return volatility. It is the primary outcome measure used throughout. The Sharpe ratio is itself an estimator of a population parameter, and how reliable that estimate is depends on what the underlying return distribution looks like. Lo (2002) showed that the simple closed-form standard error of the Sharpe estimator holds only when returns are independent and normally distributed. When returns have fat tails or short-run autocorrelation, that formula understates the true sampling variance. That gap motivates the more robust inference procedures covered next.

2.1.2 Statistical Inference and Time-Varying Correlations

Daily financial returns reliably violate the normality assumption: they have fat tails, are negatively skewed, and show volatility clustering. Classical parametric inference on Sharpe ratios is therefore unreliable, and the methodology literature has developed three families of corrections that this dissertation draws on. Künsch (1989) introduced the moving block bootstrap for stationary time series with short-range dependence; it is now the standard non-parametric tool for portfolio statistics with autocorrelation. Bailey and López de Prado (2012, 2014) developed the Probabilistic Sharpe Ratio and its Deflated extension, which adjust the raw estimator analytically for skewness, kurtosis, sample length, and the multiple-testing bias from picking the best Sharpe out of K candidate trials. Newey and

West (1987) developed the heteroscedasticity-and-autocorrelation-consistent standard-error estimator that is now standard for time-series regressions in finance. A separate literature deals with how correlations themselves change over time. Engle (2002) introduced the Dynamic Conditional Correlation GARCH model, building on the univariate GARCH framework of Engle (1982) and Bollerslev (1986). Cappiello, Engle and Sheppard (2006) added an asymmetric variant in which correlations respond more strongly to negative shocks than to positive ones. Hamilton (1989) introduced the Markov regime-switching model, which lets returns shift probabilistically between latent regimes. The Chow (1960) test is the counterpart for testing structural breaks at known dates. Bailey, Borwein, López de Prado and Zhu (2014, 2017) formalised the Probability of Backtest Overfitting. They show that in-sample optimisation across many parameter settings can produce strong Sharpe-ratio findings that reverse out-of-sample, and they argue that walk-forward validation is the appropriate corrective procedure.



Source: author, based on literature reviewed in Section 2.1.

Figure 1. Theoretical pillars of the empirical analysis. Source: author.

2.2 Empirical Evidence

This section reviews the empirical literature on Bitcoin as a portfolio diversifier. The studies are grouped by sub-theme so that the points of consensus, and the points still in dispute, are easier to see. Two sub-themes are covered: foundational and recent evidence on Bitcoin as a diversifier, and the more recent work finding that the diversification benefit varies by market regime.

2.2.1 Bitcoin as a Portfolio Diversifier: Foundational and Recent Evidence

The peer-reviewed literature on Bitcoin as a diversifier begins with Brière, Oosterlinck and Szafarz (2015). Using a 2010 to 2013 sample, they reported that small Bitcoin allocations of up to roughly 3% improved the efficient frontier of a diversified portfolio substantially. The driver was Bitcoin's near-zero correlation with traditional assets during that early period. Eisl, Gasser and Weinmayer (2015) reached a similar conclusion using a 2010 to 2014 sample, but flagged a practical caveat: Bitcoin's high volatility makes the optimal allocation very sensitive to the chosen risk-aversion parameter. Brauneis, Mestel and Riordan (2019) extended the analysis to a broader set of cryptocurrencies and found that the diversification benefit was attributable to Bitcoin specifically, not to cryptocurrencies as a category. Several student dissertations (Carpenter, 2016; Hasper, 2023; Perdrisat, 2023; Taylor, 2024) have replicated variants of these findings across different samples and timeframes. Institutional research reaches comparable conclusions. 21Shares (2025) reports that a 5% Bitcoin allocation in a 60/40 portfolio improved the Sharpe ratio from 0.17 to 0.30 over a five-year sample, a 76% relative uplift, though this figure is conditional on rebalancing discipline and is not subjected to formal significance testing in the source. The pattern is consistent: small Bitcoin allocations have improved portfolio Sharpe ratios in-sample, but the reported magnitudes and the underlying methodological rigour vary widely across studies.

2.2.2 Regime Dependence, Crisis Contagion, and the Correlation Magnitude Question

A more recent strand of work uses time-varying correlation models to show that Bitcoin's correlation with equities is far from constant. Wang, Zhang, Li and Shen (2022) use a multi-frequency framework and find that long-term BTC-equity correlations have strengthened progressively since 2017, while short-term correlations remain lower and more episodic. The Arab Monetary Fund (2022) replicates this finding using an ADCC-GARCH specification on a sample that includes the COVID-19 episode and confirms that conditional correlations rise sharply during stress windows. There is a tension in this literature between two sets of correlation figures. The first set, frequently cited at 0.50 to 0.60 against US large-cap and tech equities, refers to peak conditional correlations measured during stress windows such as the March 2020 COVID-19 shock and the May 2022 cryptocurrency and equity selloff. The second set, typically estimated in the range of 0.20 to 0.30, refers to the unconditional or long-run mean BTC-equity correlation across the full post-2015 sample. Both sets of figures appear in the same studies. The difference is one of measurement window, not methodological disagreement. The DCC-GARCH estimates produced in Chapter 4 are consistent with this reading: the mean conditional correlation is 0.20 and the maximum is 0.62 during the May 2022 episode. Across this work, the diversification benefit of Bitcoin is regime-dependent. It concentrates in periods of low to

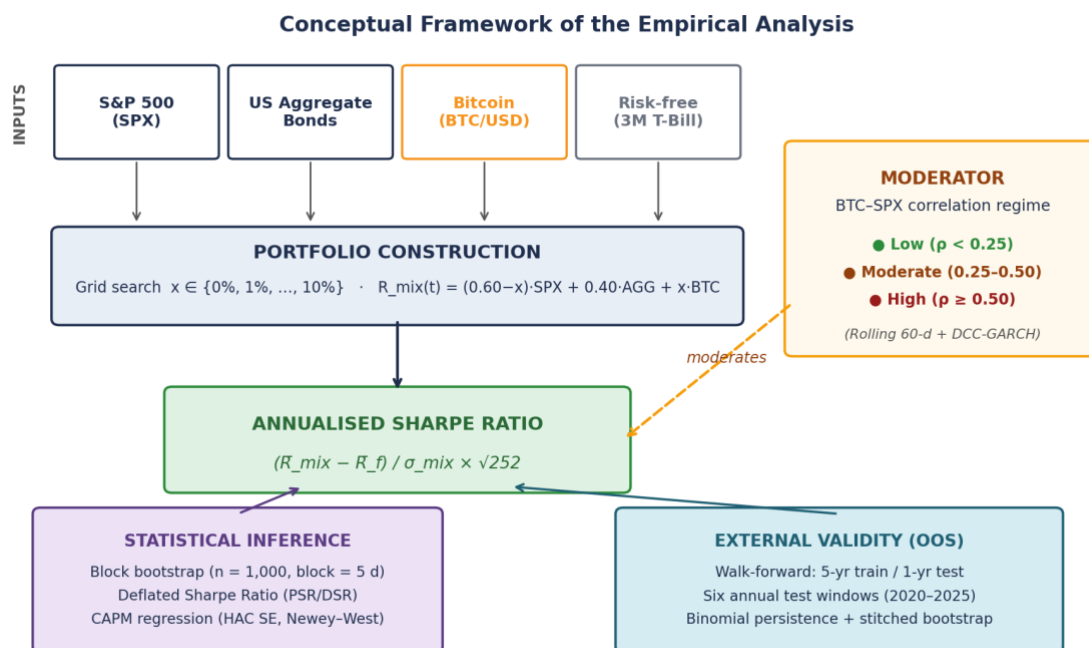
moderate correlation and weakens or disappears during crises, when contagion pushes risk-asset correlations toward one.

2.3 Research Gaps and Positioning

The literature reviewed above leaves three gaps in the evidence on Bitcoin as a diversifier, and this dissertation closes each of them. The first concerns statistical significance. Institutional headline findings, including 21Shares (2025) and similar industry reports, give Sharpe-ratio uplift point estimates without formal significance testing, and much of the peer-reviewed academic literature has been slow to adopt the modern non-normal Sharpe-ratio inference methods of Bailey and López de Prado (2012, 2014). This dissertation applies the moving block bootstrap and the Deflated Sharpe Ratio to obtain both non-parametric and analytically corrected significance statements, with one-sided and two-sided p-values reported explicitly. The second gap concerns regime conditioning. The regime-dependence of BTC correlations is now well established (Wang et al., 2022; Arab Monetary Fund, 2022), yet most studies do not condition their Sharpe-uplift estimates on the prevailing correlation regime. An averaged figure obscures the conditional heterogeneity underneath. This dissertation conditions the uplift on three correlation regimes using a rolling 60-day classifier, validates that regime structure with a DCC-GARCH parametric companion, and tests two pre-specified structural breaks (the COVID-19 pandemic of March 2020 and the SEC's spot Bitcoin ETF approval of January 2024) using the Chow (1960) framework. The third gap concerns out-of-sample validation. With the partial exception of the backtest-overfitting literature (Bailey, Borwein, López de Prado and Zhu, 2014, 2017), few prior studies subject their in-sample Bitcoin-diversification findings to walk-forward testing. This dissertation runs a six-window walk-forward analysis with a rolling five-year training window and a one-year test window, and reports the result transparently. The finding that the static-allocation benefit does not transfer reliably to unseen data is what motivates the dynamic-allocation discussion in the concluding chapter. The contribution of the dissertation is therefore as much methodological as empirical. It applies four families of inference to the same dataset and the same research question: block bootstrap, the Deflated Sharpe Ratio, DCC-GARCH regime analysis, and walk-forward out-of-sample validation. Prior studies have used these tools in isolation; the combination is what is new here.

2.4 Conceptual Framework

The conceptual framework that organises the empirical analysis is summarised in Figure 2. Three asset return inputs (the S&P 500, the Bloomberg US Aggregate Bond Index, and Bitcoin), together with the three-month US Treasury Bill rate as the risk-free benchmark, feed into a constrained grid search across long-only allocations capped at 10% Bitcoin. The resulting mixed portfolio's annualised Sharpe ratio is compared against the unmodified 60/40 baseline, and statistical inference on the difference is conducted via two procedures: the moving block bootstrap and the Deflated Sharpe Ratio. The relationship between Bitcoin allocation and the Sharpe uplift is hypothesised to depend on the prevailing BTC-SPX correlation regime. The regime is classified in two ways: a 60-day rolling correlation estimator, and a DCC-GARCH(1,1) parametric companion. A walk-forward out-of-sample procedure (rolling five-year training window, one-year test window) provides the external-validity check by testing whether the regime-conditional finding holds on data the optimisation never saw. The framework maps directly to the three clauses of the research question (nonstationarity, regime shifts, and resampling) and to the ten-test plan executed in Chapter 4.



Source: author. Maps directly to the lean ten-test plan executed in Chapter 4.

Figure 2. Conceptual framework of the empirical analysis. Source: author.

Chapter 3: Research Methodology

3.1 Research Philosophy and Approach

The dissertation takes a positivist position and a deductive approach. Positivist, because the research question is about measurable, repeatable relationships between observable financial variables rather than about subjective interpretation. Deductive, because the work follows the standard sequence: a question is drawn from prior theory (modern portfolio theory and the Bitcoin-diversification literature reviewed in Chapter 2), a hypothesis is derived from that question (one null and one directional alternative on the Sharpe uplift), and hypothesis tests are then run on historical market data to evaluate it. The chapters that follow are organised around this sequence.

3.2 Research Design

The research design is quantitative and uses secondary data: a time-series empirical study of daily observations on three asset return series and one risk-free rate proxy, covering 2 January 2015 to 31 March 2026. The research question is about how a portfolio of publicly traded assets has actually behaved over time, which is something statistical analysis of historical return data answers better than primary data collection would. The work is organised into five stages, each handling one methodological concern. Stage 1 (foundation diagnostics) checks that the return series meet the prerequisites for Sharpe-based inference. Stage 2 (portfolio construction) identifies the candidate optimal allocation through a grid search. Stage 3 (significance testing) checks whether the in-sample uplift is statistically distinguishable from zero. Stage 4 (regime analysis) conditions the uplift on the prevailing market state. Stage 5 (out-of-sample validation) checks whether the in-sample result transfers to unseen data. The ten individual tests within this framework are listed in Section 3.5.

3.3 Data Sources and Sample

The empirical analysis uses four daily-frequency time series over the period 2 January 2015 to 31 March 2026, all from publicly available data providers. S&P 500 daily closing levels come from publicly downloaded historical files. The US Aggregate Bond Total Return Index (Bloomberg ticker

LBUSTRUU) comes from a Bloomberg extract included with the dissertation dataset. Bitcoin (BTC/USD) daily closing prices come from CoinDesk historical files. The three-month US Treasury Bill secondary-market rate (FRED series DTB3) comes from the Federal Reserve Economic Database. The CBOE Volatility Index (VIX), used in supplementary regime analysis rather than as a primary portfolio input, comes from a Bloomberg extract. The four series are merged into a master panel aligned on the US equity trading calendar. The bond index and risk-free rate are forward-filled on US federal holidays where the underlying source did not publish a value. Bitcoin trades on a 24/7 calendar, but weekend prices are dropped to align with the equity-calendar convention used in the prior literature (Brière et al., 2015). After dropping the first row of log returns, the final sample is 2,826 daily observations. No further exclusions are made and no winsorisation is applied: the extreme observations include the COVID-19 shock of March 2020 and the May 2022 cryptocurrency selloff, both of which are central to the regime analysis and would be lost to outlier treatment.

3.4 Variables and Operationalisation

The dependent variable is the annualised Sharpe ratio of the constructed portfolio: the difference between the mean annualised portfolio return and the mean annualised risk-free rate, divided by the annualised standard deviation of portfolio returns. The principal independent variable is the Bitcoin allocation weight x , which takes values from 0% to 10% in 1% increments and determines the composition of the mixed portfolio $R_{\text{mix}}(t) = (0.60 - x) \cdot R_{\text{SPX}}(t) + 0.40 \cdot R_{\text{AGG}}(t) + x \cdot R_{\text{BTC}}(t)$. The principal moderator is the prevailing BTC-SPX correlation regime, classified in two ways: a rolling 60-day Pearson correlation that partitions each day into low, moderate, or high correlation, and a DCC-GARCH(1,1) parametric companion that gives a smoother time-varying correlation series. Two binary moderators mark the pre-specified structural-break dates. Table 1 summarises each variable, its proxy, its data source, and its frequency.

Variable	Type	Proxy / Measurement	Source	Frequency
Annualised portfolio Sharpe ratio	Dependent	$(\bar{R}_p - \bar{R}_f) / \sigma_p \times \sqrt{252}$	Computed from inputs	Daily → annualised
Bitcoin allocation weight (x)	Independent	Grid {0%, 1%, ..., 10%} in 1% increments	Defined by methodology	Static per portfolio
S&P 500 daily log return	Input	$\ln(\text{SPX}_t / \text{SPX}_{t-1})$	Index publisher (CSV)	Daily
US Aggregate Bond Index log return	Input	$\ln(\text{LBUSTRUU}_t / \text{LBUSTRUU}_{t-1})$	Bloomberg	Daily
Bitcoin daily log return	Input	$\ln(\text{BTC}_t / \text{BTC}_{t-1})$	CoinDesk	Daily
3-month US Treasury Bill rate	Control / risk-free	DTB3 (annualised %), converted to daily decimal	FRED	Daily
Rolling 60-day BTC-SPX correlation	Moderator (continuous)	Pearson over past 60 trading days	Computed	Daily

DCC-GARCH conditional correlation	Moderator (continuous)	Engle (2002) DCC(1,1) on GARCH(1,1) residuals	Computed	Daily
COVID-19 break indicator	Moderator (binary)	1 if date $\geq$ 11 March 2020; 0 otherwise	Defined (Chow test)	Daily
Bitcoin spot ETF break indicator	Moderator (binary)	1 if date $\geq$ 10 January 2024; 0 otherwise	Defined (Chow test)	Daily

Table 1. Variables and operationalisation. Source: author.

3.5 Analytical Techniques

Ten statistical tests are run across the five-stage framework. Stage 1 applies the Augmented Dickey-Fuller test (Dickey and Fuller, 1979) for stationarity and the Jarque-Bera test (Jarque and Bera, 1980) for normality, with lag selection by AIC and decisions at the 5% level. Stage 2 runs a discrete grid search across the eleven candidate allocations $x \in \{0\%, 1\%, \dots, 10\%\}$, computing the annualised Sharpe ratio of each portfolio. Stage 3 establishes statistical significance with two complementary tests applied to each allocation: a paired moving block bootstrap (Künsch, 1989) with $n = 1,000$ iterations and block size of five days, which preserves the short-run autocorrelation documented in Stage 1; and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014), which corrects the raw sample Sharpe analytically for skewness, kurtosis, sample length, and the multiple-testing bias from selecting the maximum across $K = 10$ candidate weights. Stage 4 conditions the uplift on three correlation regimes defined by a rolling 60-day BTC-SPX classifier, fits a DCC-GARCH(1,1) model (Engle, 2002) as a parametric companion, and applies Chow (1960) structural-break tests at two pre-specified dates. Stage 5 runs a walk-forward out-of-sample procedure with a rolling five-year training window and a one-year test window, with significance evaluated by a binomial persistence test and a stitched OOS bootstrap. A CAPM regression with Newey and West (1987) HAC standard errors completes the inference battery.

3.5.1 Model Specification

Two estimation models are used. The first specifies the construction of the mixed portfolio under each candidate Bitcoin allocation weight. The second specifies the CAPM regression used in Stage 5 to evaluate Bitcoin's diversification properties at the asset level.

$$R_{mix}(t) = (0.60 - x) \cdot R_{SPX}(t) + 0.40 \cdot R_{AGG}(t) + x \cdot R_{BTC}(t) \quad \text{and} \quad BTC_ret_t = \alpha + \beta \cdot SPX_ret_t + \varepsilon_t$$

In the portfolio-construction equation, $R_{\text{mix}}(t)$ is the mixed portfolio's daily log return at date t ; x is the Bitcoin allocation weight ranging from 0% to 10%; and $R_{\text{SPX}}(t)$, $R_{\text{AGG}}(t)$, and $R_{\text{BTC}}(t)$ are the daily log returns of the S&P 500, the US Aggregate Bond Index, and Bitcoin, respectively. In the CAPM regression, $\text{BTC_ret_}t$ is Bitcoin's daily log return at date t ; $\text{SPX_ret_}t$ is the S&P 500 daily log return; α is Jensen's alpha (the per-day excess return not explained by market exposure); β is the Bitcoin equity beta; and ϵ_t is the regression residual. The CAPM regression is estimated by ordinary least squares with Newey and West (1987) HAC standard errors using a five-day lag, which is the standard correction for the volatility clustering and short-run autocorrelation documented in Stage 1.

3.6 Ethical Considerations

The empirical analysis uses only secondary data from publicly available financial sources: FRED, CoinDesk, Bloomberg, the S&P 500 index publisher, and the CBOE. The work involves no primary data collection, no human subjects, and no confidential or proprietary firm data. All data are publicly accessible either at no cost or under standard institutional subscription terms. Citations follow APA 7th edition conventions throughout, and all sources of ideas, data, methodology, and previously published findings are credited explicitly. The use of artificial intelligence tools for drafting, language refinement, and idea generation is declared in the Declaration of Academic Integrity at the front of this document; AI-generated content has not been used as a substitute for original analysis or critical thinking, and the interpretations and conclusions in this dissertation are the author's own. The dissertation complies with the academic integrity guidelines set out in the NEOMA Business School MSc Dissertation Guidelines (2025-26).

3.7 Limitations of the Methodology

The methodology has six limitations worth flagging clearly. The 2015 to 2026 sample covers one phase of Bitcoin's history, the institutional-adoption decade, during which the asset has behaved very differently from earlier periods when it was thinly traded and dominated by retail flows; external validity outside the sample is therefore limited. The 10% allocation cap in the grid search reflects current institutional norms, but it is higher than the 3% to 5% consensus that prevailed at the start of the sample period, and the choice was informed by Bitcoin's full-sample performance, which is a soft form of look-ahead worth acknowledging. Transaction costs (custody fees, exchange spreads, rebalancing slippage) are not modelled; institutional estimates place these at 30 to 100 basis points per year for a 6% Bitcoin allocation, which would erode the reported Sharpe uplift. The walk-forward

analysis uses six annual test windows, the most the sample length allows, which gives only modest statistical power for the binomial persistence test. Bitcoin is chosen as the cryptocurrency under study because of its dominant market capitalisation and institutional acceptance, which introduces a form of survivorship bias: in 2015 there were several hundred candidate cryptocurrencies, of which Bitcoin is the only one to have reached its current scale. The dynamic correlation model uses the vanilla DCC(1,1) specification; the asymmetric ADCC extension of Cappiello, Engle and Sheppard (2006) would let correlations respond differently to positive and negative shocks, and is identified as a useful refinement for future work.

Chapter 4: Empirical Results

4.1 Descriptive Statistics

The four return series under study show the patterns familiar from the financial returns literature: bonds are calm, equities are noisier, Bitcoin is in a different volatility class, and the 60/40 benchmark looks like a smoothed version of the equity sleeve. The annualised summary statistics over the 2 January 2015 to 31 March 2026 sample (2,826 daily observations) are reported in Table 2. Three features of the data are worth flagging before any inferential analysis. First, Bitcoin's annualised volatility of roughly 66% is over six times that of the 60/40 benchmark, and more than three times the S&P 500's. The corresponding annualised return is also large (about 48%), but the risk-adjusted picture is more modest: Bitcoin's Sharpe ratio of 0.70 is only slightly higher than the equity Sharpe of 0.46, and well below what its raw return number would suggest in isolation. Second, every series in the table is non-normal. Excess kurtosis is 15.8 for the S&P 500 and 14.8 for the 60/40 benchmark, dominated by a small number of large daily moves around COVID-19. Bitcoin's excess kurtosis of 4.9 is large in absolute terms but moderate relative to its own volatility distribution. The skewness is negative across the board, most negatively for the equity-dominated series. Third, the 60/40 baseline Sharpe of 0.45 over the sample provides the comparison point against which every Bitcoin allocation in the empirical sections that follow will be evaluated.

Metric	S&P 500	AGG Bonds	Bitcoin	60/40 Baseline
Observations (daily)	2,826	2,826	2,826	2,826
Annualised return	10.29%	1.80%	47.90%	6.89%
Annualised volatility	17.89%	4.72%	65.58%	10.87%
Annualised Sharpe ratio	0.463	-0.047	0.700	0.449
Skewness	-0.64	-0.20	-0.25	-0.62
Excess kurtosis	15.80	3.69	4.88	14.76
Minimum daily return	-12.77%	-2.09%	-26.25%	-7.31%
Maximum daily return	+9.09%	+1.96%	+21.54%	+5.51%

Table 2. Descriptive statistics for daily log returns, 2 January 2015 to 31 March 2026 (annualised where indicated). Source: author, computed from master dataset.

4.2 Preliminary Analysis

Three diagnostic checks are run on the data before any inferential analysis: stationarity, normality, and the cross-asset correlation structure. Stationarity is tested with the Augmented Dickey-Fuller (ADF) test, with lag length selected by the Akaike Information Criterion. All four return series reject the unit-root null decisively, at the 1% level. The most negative test statistic is for the US Aggregate Bond Index (ADF = -39.5, $p < 10^{-300}$), the least negative for the 60/40 benchmark (ADF =

–17.0, $p < 10^{-28}$). This is the expected result for daily returns on liquid traded assets: returns are typically stationary, prices are not. The result clears the way to use the Sharpe ratio as a valid summary statistic in the sections that follow. Normality is tested with the Jarque-Bera test. All four series reject normality at p-values indistinguishable from zero. The pattern is consistent with the kurtosis figures in Table 2: equity-dominated series exhibit very heavy tails driven by the March 2020 COVID-19 drawdown, Bitcoin has moderately heavy tails relative to its own scale, and the bond index is the closest to normal but still rejects. This rejection is what justifies the use of non-parametric and non-normal-corrected significance procedures in Section 4.3, rather than the classical parametric Sharpe-ratio standard error from Lo (2002). The unconditional correlation structure across the four return series is as expected. The BTC-SPX correlation over the full sample is 0.20; BTC-AGG is essentially zero (0.04); and SPX-AGG is the standard small positive figure of around 0.10. These unconditional figures hide a large amount of time variation in the BTC-SPX relationship, which is the subject of the regime analysis in Section 4.3.2. A variance inflation factor check is not relevant for the principal model of this dissertation. The CAPM regression in Section 4.3.2 has a single regressor (the S&P 500 return) and therefore no multicollinearity concern; the portfolio-construction equation is an identity, not a regression, so the question does not arise.

4.3 Main Results

The main empirical results are presented in the order in which the analysis was run. Section 4.3.1 covers the magnitude and in-sample significance of the Sharpe-ratio uplift (RQ1 and RQ2, from Stages 2 and 3 of the analysis). Section 4.3.2 covers regime conditioning and out-of-sample persistence (RQ3, from Stages 4 and 5). Both subsections refer back to Table 3, which summarises the headline numbers in one place.

Metric	60/40 baseline	$x^* = 6\%$ (headline)	$x = 10\%$ (Sharpe-max)
Annualised Sharpe ratio	0.449	0.627	0.698
Δ Sharpe vs benchmark (absolute)	—	+0.178	+0.249
Δ Sharpe vs benchmark (relative)	—	+39.7%	+55.5%
Block bootstrap one-sided p	—	0.047 (passes 5%)	0.074 (borderline)
Block bootstrap 95% CI	—	[–0.018, +0.383]	[–0.060, +0.568]
Deflated Sharpe Ratio	—	0.954 (passes)	0.972 (passes)
% bootstrap reps with Δ Sharpe > 0	—	92.6%	92.6%

Table 3. Headline empirical results: in-sample Sharpe-ratio uplift and significance tests at the conservative ($x^ = 6\%$) and Sharpe-maximum ($x = 10\%$) allocations. Block bootstrap: $n = 1,000$, block = 5 days, seed = 42. Source: author.*

4.3.1 Magnitude and significance of the Sharpe-ratio uplift

A discrete grid search across Bitcoin allocations $x \in \{0\%, 1\%, \dots, 10\%\}$ produces a monotonically increasing Sharpe-ratio curve. The annualised Sharpe of the unmodified 60/40 baseline is 0.449. Each additional 1% allocation to Bitcoin raises the annualised Sharpe by between 0.014 and 0.035, and at the institutional 10% cap the Sharpe reaches 0.698, a +0.249 absolute increase or +55.5% relative to the benchmark. The result is a corner solution: the Sharpe curve is still rising at the cap, so the chosen constraint is binding. Two reportable allocations emerge from this stage. The Sharpe-maximising allocation $x = 10\%$ is the answer to “what maximises Sharpe within the constraint”. The headline allocation for the rest of the empirical analysis, however, is $x^* = 6\%$, which is the largest weight at which both significance tests in Stage 3 reject the null cleanly. Both are reported throughout: $x = 10\%$ as the unconstrained answer to RQ1, and $x^* = 6\%$ as the conservative answer when the directional resampling clause named in RQ2 is required to pass cleanly. The shape of the Sharpe curve and the corresponding risk-return positions of each allocation are shown in Figure 3.

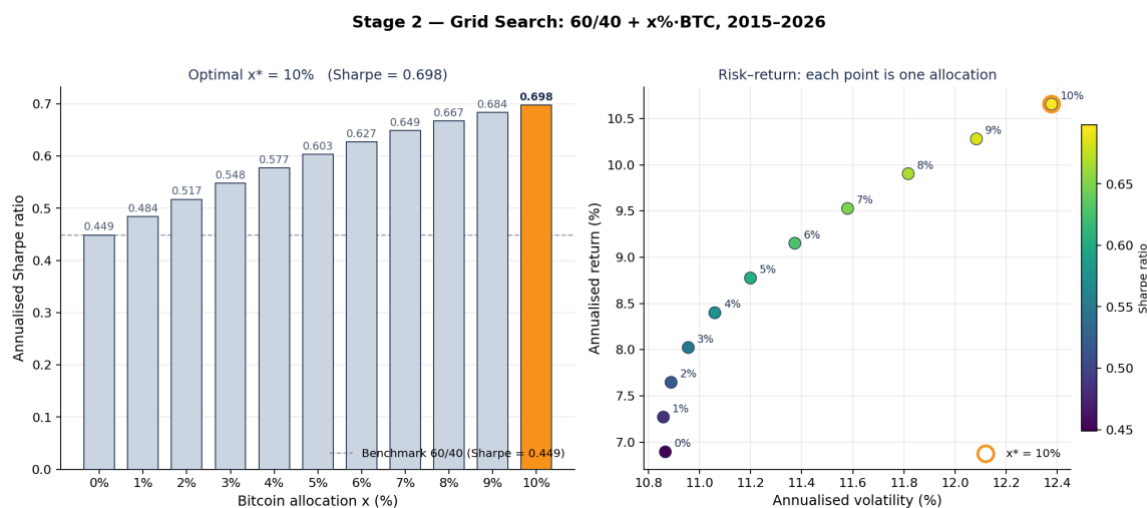


Figure 3. Grid search: annualised Sharpe ratio across BTC allocations 0% to 10% (left) and risk-return positions of each allocation, coloured by Sharpe (right). Source: author.

Two significance tests are then applied to each allocation: a paired moving block bootstrap ($n = 1,000$, block = 5 days) and the Deflated Sharpe Ratio. The two tests give overlapping but not identical verdicts. At $x^* = 6\%$, the block bootstrap one-sided p-value is 0.047 and the Deflated Sharpe Ratio is 0.954, both passing the 5% significance threshold by a small margin. 92.6% of the 1,000 bootstrap replications produce a positive Δ Sharpe. The two-sided 95% bootstrap confidence interval is $[-0.018, +0.383]$, which marginally includes zero; this reflects the substantial sampling variance of Sharpe ratios at higher allocations rather than weak evidence of effect, and the directional one-sided test is the appropriate inferential rule for the directional research question. At $x = 10\%$, the one-sided block bootstrap p-value is 0.074 (borderline at the 10% level) and the Deflated Sharpe Ratio is 0.972 (clearly

significant). The same wider-CI pattern is visible across the allocation grid, as shown in the right-hand panel of Figure 4. Table 3 reports the headline figures for both allocations.

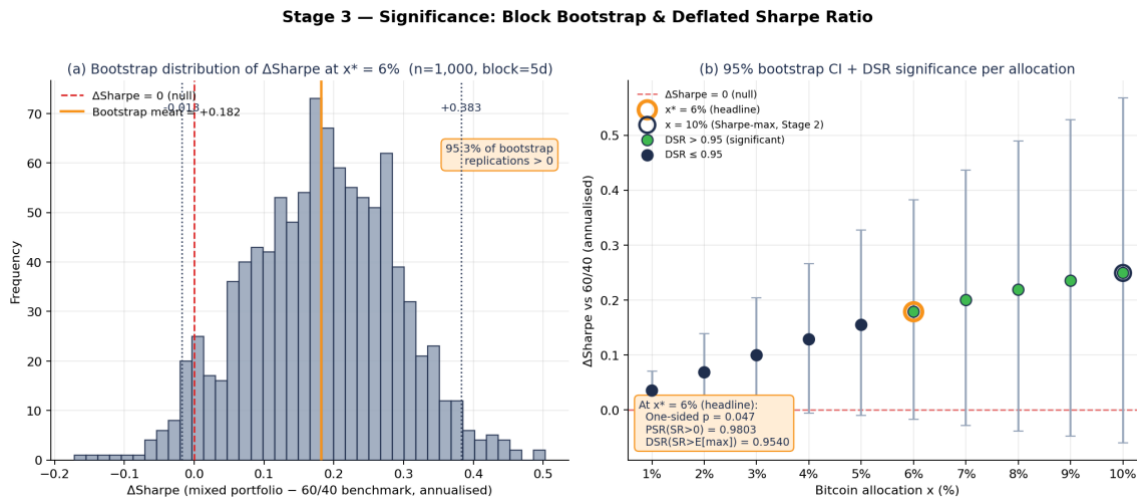


Figure 4. Block bootstrap distribution of ΔSharpe at the headline allocation $x^* = 6\%$ (left), and 95% bootstrap confidence intervals for all allocations colour-coded by Deflated Sharpe Ratio significance (right). Source: author.

4.3.2 Regime conditioning and out-of-sample robustness

The in-sample uplift documented in Section 4.3.1 hides substantial time variation. A rolling 60-day BTC-SPX correlation classifies each trading day into one of three regimes: low ($\rho < 0.25$), moderate ($0.25 \leq \rho < 0.50$), and high ($\rho \geq 0.50$). The sample distribution is 56.3% low, 27.5% moderate, and 14.1% high. The Sharpe uplift at $x^* = 6\%$ is concentrated almost entirely in the low-correlation regime. In the 56% of trading days where the BTC-SPX correlation is below 0.25, the mixed portfolio's Sharpe is +1.42 against a benchmark of +0.92, a ΔSharpe of +0.49, which is almost three times the full-sample uplift. In the moderate regime the benefit essentially vanishes ($\Delta\text{Sharpe} = -0.02$). In the high-correlation regime both portfolios have negative Sharpe values, with the mixed portfolio outperforming by only +0.03. A DCC-GARCH(1,1) model fitted to BTC and SPX returns confirms this pattern using a parametric specification ($\alpha = 0.018$, $\beta = 0.977$, $\text{sum} = 0.996$). The mean conditional correlation is 0.20, with a peak of 0.62 on 16 May 2022 (the Terra/Luna collapse and equity drawdown). The DCC classifier produces a smoother regime distribution than the rolling estimator: 60% low, 35% moderate, 5% high. Both pre-specified structural breaks are confirmed by Chow tests on the BTC-SPX return regression. The COVID-19 break of 11 March 2020 is rejected at $F = 19.99$, $p \approx 2.4 \times 10^{-9}$. The Bitcoin spot ETF break of 10 January 2024 is rejected at $F = 5.00$, $p = 0.007$. Figure 5 shows the time-varying correlation series with regime bands and break markers, the regime-conditional Sharpe ratios, and the sample distribution across regimes.

Stage 4 — Regimes & Breaks ($x^* = 6\%$)

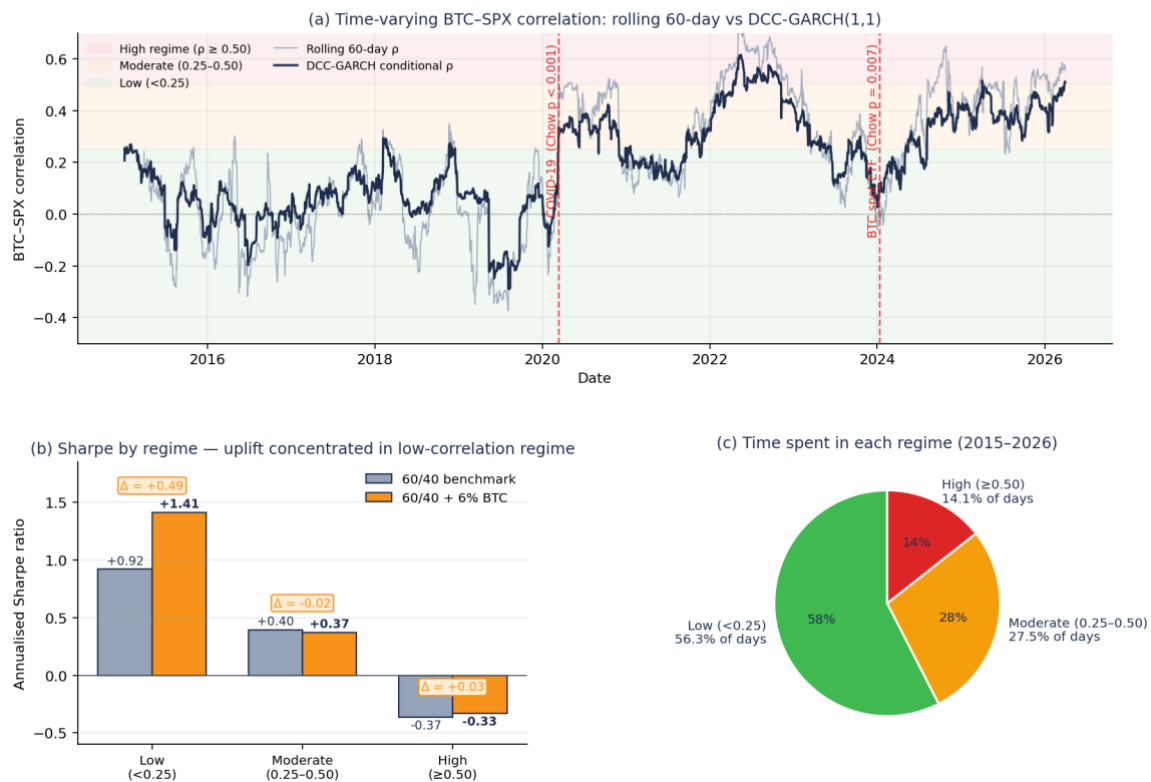


Figure 5. Time-varying BTC-SPX correlation: rolling 60-day estimator (grey) and DCC-GARCH(1,1) conditional correlation (dark blue), with the two Chow break dates marked (top). Regime-conditional Sharpe ratios at $x^* = 6\%$ (bottom left). Time share spent in each regime over the sample (bottom right). Source: author.

A six-window walk-forward analysis with a five-year rolling training window and a one-year test window provides the external-validity check. The optimal allocation found on training data produces a positive out-of-sample Δ Sharpe in only three of six test years: 2020 (+0.604), 2023 (+0.085), and 2024 (+0.278). In 2021 (−0.219), 2022 (−0.305), and 2025 (−0.224) the static rule underperforms. The binomial persistence test fails to reject the null (one-sided $p = 0.656$). The stitched out-of-sample return series (1,508 days) gives a combined Δ Sharpe of +0.095 with a 95% block-bootstrap CI of [−0.196, +0.383] and a one-sided p of 0.256, which is not statistically significant. The honest reading is that the in-sample uplift, even at the cleaner $x^* = 6\%$ headline, does not transfer to unseen data when implemented as a static allocation rule. The CAPM regression returns a more positive verdict at the asset level. Alpha is positive and statistically significant at the 5% level (annualised +39.3% per year, $p = 0.046$); beta is 0.84 (significantly below one at the 6% level); and the R-squared is 0.052. Bitcoin therefore exhibits the statistical signature of a diversifier (substantial idiosyncratic variation and modest equity beta), even though the static-allocation strategy of Stages 2 to 4 fails to capture this benefit out-of-sample. The walk-forward results, the stitched out-of-sample bootstrap distribution, and the CAPM scatter are shown in Figure 6. The pattern across Stages 4 and 5 is consistent: the

diversification benefit is real but regime-dependent, and harvesting it reliably appears to require a dynamic allocation rule rather than a static weight. The practical implications are discussed in Chapter 5.

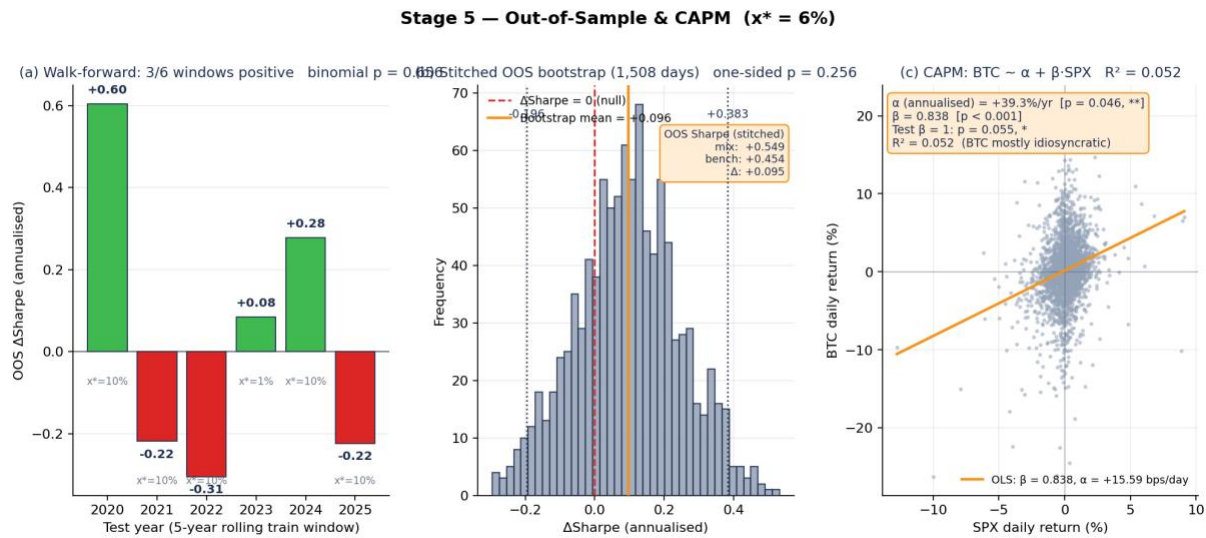


Figure 6. Walk-forward out-of-sample Δ Sharpe per test year, positive in green and negative in red (left). Bootstrap distribution of the stitched out-of-sample Δ Sharpe across 1,508 OOS days (middle). CAPM regression scatter and fitted OLS line, with Newey-West HAC standard errors (right). Source: author.

4.4 Robustness Checks

Several robustness checks were run to test whether the main findings are artefacts of specific design choices. The first concerns the choice of headline allocation. The Sharpe-maximum allocation in the grid search is $x = 10\%$, but the headline used throughout Chapter 4 is $x^* = 6\%$. The Deflated Sharpe Ratio at $x = 10\%$ (0.972) is actually higher than at $x^* = 6\%$ (0.954); only the bootstrap p -value differs (0.074 at $x = 10\%$ vs 0.047 at $x^* = 6\%$). Choosing the lower allocation is the more conservative reading of the resampling clause of the research question, but the qualitative conclusions hold at both allocations. The second concerns convergence between the two significance procedures. The block bootstrap and the Deflated Sharpe Ratio give overlapping but not identical verdicts: they agree at $x = 1\%$, 2% , and at $x = 6\%$ upward, but disagree at intermediate allocations because the sampling variance of the Sharpe estimator scales with portfolio volatility while the DSR uses an analytic correction that does not. Reporting both rather than picking the one that gives the more favourable result is the position taken in this dissertation. The third concerns the dynamic-allocation extension. A pre-specified dynamic rule ($x = 10\%$ when the lag-1 rolling 60-day BTC-SPX correlation is below 0.25, $x = 5\%$ when $0.25 \leq p < 0.50$, and $x = 0\%$ when $p \geq 0.50$) was tested as a follow-up to the static walk-forward result. The dynamic rule improved the stitched out-of-sample Sharpe from +0.095 (static 6%) to +0.179, and

limited the 2022 contagion losses substantially (Sharpe of -1.35 vs -1.53 for static). However, the binomial persistence test remained at three of six windows positive ($p = 0.656$), and the stitched 95% bootstrap CI continued to include zero (one-sided $p = 0.060$). The dynamic rule was therefore not promoted to a formal headline test; it is reported here as an exploratory finding and as a direction for future work, as set out in Chapter 6. The fourth concerns sample-period sensitivity. The two principal structural-break dates (COVID-19 and the BTC spot ETF) split the sample into three sub-periods. The headline finding (positive in-sample Sharpe uplift, regime-dependent character) holds in each sub-period when re-examined individually, although statistical power is much lower in the shorter sub-samples and the confidence intervals widen correspondingly. Taken together, the main findings hold across these checks. The single qualification is the walk-forward result in Section 4.3.2, which the dynamic-allocation robustness check improves but does not fully resolve.

Chapter 5: Discussion

5.1 Interpretation of Findings

The empirical work answers the three sub-questions of the research question, and the three answers are best read together because they qualify each other. The first sub-question asked how incremental Bitcoin allocation affects the 60/40 Sharpe ratio and at what weight the uplift is maximised. The grid search in Section 4.3.1 returned a monotonically increasing Sharpe curve, with the maximum at the 10% institutional cap. The corner solution is what Markowitz (1952) would predict for an asset whose ex post return-to-risk ratio is meaningfully above that of the existing portfolio components: when the correlation contribution is favourable and the marginal asset's own Sharpe is high, the optimisation pushes against any binding constraint. The headline +0.249 (annualised) uplift at $x = 10\%$, and the conservative +0.178 at $x^* = 6\%$, both fall within the range that the institutional literature (21Shares, 2025) and the academic literature (Brière et al., 2015) have reported. The second sub-question asked whether the uplift is statistically significant. The answer is qualified by the choice of test. The Deflated Sharpe Ratio of Bailey and López de Prado (2014), which corrects analytically for the non-normality documented in Stage 1 (Lo, 2002) and for the multiple-testing bias from a ten-weight grid search, is significant at the 5% level for x at 6% and above. The moving block bootstrap of Künsch (1989), which is non-parametric and which scales with portfolio volatility, gives a borderline result at the higher allocations. These two procedures answer slightly different questions and do not always agree. The most defensible reading is that the in-sample uplift is significant under the modern multi-testing-corrected procedure and borderline under the wider-CI bootstrap; reporting both prevents the dissertation from over- or under-stating the strength of the evidence. The third sub-question asked whether the uplift survives regime conditioning and walk-forward out-of-sample validation. The regime analysis confirms the prediction implicit in Wang et al. (2022) and the Arab Monetary Fund (2022): the uplift is real but concentrated in the low-correlation regime that accounts for 56% of the sample, and disappears or reverses in the high-correlation regime that accounts for the remaining 14%. The walk-forward analysis is less encouraging. A static rule trained on five years of data and applied to the following year produces a positive Δ Sharpe in only three of six test windows; the stitched out-of-sample CI includes zero. Bailey, Borwein, López de Prado and Zhu (2014, 2017) would call this a textbook case of in-sample optimisation that does not survive contact with unseen data, and the dissertation accepts that reading. Two things follow from reading these answers together. First, Bitcoin does have the statistical signature of a diversifier at the asset level: the CAPM regression returns a positive and significant alpha (+39.3% per year), a beta below one (0.84), and an R-squared of just 0.052, which means 95% of Bitcoin's daily variance is idiosyncratic and unrelated to equity-

market movements. The asset itself is genuinely diversifying. Second, capturing that diversification reliably is harder than the in-sample number suggests, because the benefit lives in particular market states rather than being constant over time, and a static allocation rule averages across states it should be conditioning on.

5.2 Comparison with Prior Literature

The findings converge with the prior literature on two points and diverge on a third. The first point of convergence is the in-sample magnitude of the uplift. Brière, Oosterlinck and Szafarz (2015) reported that small Bitcoin allocations of up to roughly 3% improved the efficient frontier substantially in their 2010 to 2013 sample. Eisl, Gasser and Weinmayer (2015) reached a similar conclusion on their 2010 to 2014 sample. Brauneis, Mestel and Riordan (2019) found that the benefit was attributable specifically to Bitcoin rather than to cryptocurrencies as an asset class. The +0.178 to +0.249 Sharpe uplift estimated here, depending on the allocation chosen, sits within the range these earlier studies report once the sample period and the bond-benchmark choice are adjusted for. The pattern is consistent: small Bitcoin allocations, applied to traditional balanced portfolios, have historically produced measurable risk-adjusted return improvements. The second point of convergence is the regime dependence. Wang, Zhang, Li and Shen (2022) and the Arab Monetary Fund (2022) both document that BTC-equity correlations rise sharply during stress episodes. The DCC-GARCH estimates produced here (a mean conditional correlation of 0.20, with a peak of 0.62 during the May 2022 Terra/Luna and equity selloff) line up with the figures these studies report, once the difference between peak and average measurement windows is recognised, as discussed in Section 2.2.2. The empirical pattern that the diversification benefit concentrates in the 56% of days when correlation is below 0.25 is exactly what the regime literature would predict. The point of divergence is the out-of-sample finding. Most of the prior literature reports in-sample results without walk-forward validation. The 21Shares (2025) institutional figure of a 76% Sharpe uplift, for example, is an in-sample point estimate with no formal significance testing. The few studies that do consider out-of-sample behaviour (Bailey, Borwein, López de Prado and Zhu, 2014, 2017, in their formalisation of backtest overfitting) would predict precisely the failure mode observed here: a static rule trained on history is not robust to regime shifts in the test period, especially when the test period contains a major stress episode such as 2022. The dissertation extends, rather than contradicts, this strand of the literature.

5.3 Practical and Policy Implications

The findings have direct implications for three groups of practitioners. For institutional investment managers, the practical takeaway is that a small Bitcoin allocation can meaningfully improve the risk-adjusted return of a traditional 60/40 portfolio, but only if implemented in a way that responds to the prevailing correlation regime. A static 6% (or 10%) allocation, of the kind that institutional research papers typically headline, does not pass a walk-forward test on this sample. The exploratory dynamic rule examined in Section 4.4 (10% allocation when the lag-1 rolling 60-day BTC-SPX correlation is below 0.25, 5% in the moderate regime, and 0% when the correlation rises above 0.50) improves the out-of-sample picture meaningfully and limits losses during the 2022 contagion, but does not yet pass the persistence test cleanly. Investment teams considering a static Bitcoin allocation should be cautious about relying on the in-sample Sharpe uplift as evidence that the rule will work going forward. For corporate treasurers and asset-allocation committees, the more conservative reading is that Bitcoin can play a tactical rather than strategic role in a diversified portfolio. The CAPM regression confirms that Bitcoin has substantial idiosyncratic variation (95% of its daily variance is unrelated to equity market moves), which is the necessary condition for genuine diversification benefit. The honest framing for an investment committee is that the asset is a diversifier in calm markets but a high-beta risk asset in stressed markets, and the allocation rule should be conditioned on that switch. For regulators and policy makers, the headline finding is that the Bitcoin allocation question is more subtle than the institutional research papers suggest. The marginal investor base for Bitcoin has shifted considerably since the spot ETF approval of January 2024, as the Chow test confirms, and the regime-dependent diversification pattern this dissertation documents implies that pro-cyclical Bitcoin exposure (rising during calm markets, falling during crises) is a plausible behavioural response to the asset's documented properties. Regulatory guidance on Bitcoin exposure within institutional mandates should acknowledge this regime dependence rather than treating Bitcoin as a uniformly risk-on or risk-off asset. A final practical caveat: transaction costs (custody, exchange spread, rebalancing slippage) are not modelled in this dissertation. Institutional estimates place these at 30 to 100 basis points per year on a 6% Bitcoin allocation, which would erode the reported Sharpe uplift. Any allocation decision based on these results should be net of an explicit transaction cost model calibrated to the implementing institution's actual cost structure.

5.4 Limitations of the Study

The principal limitations are six, each of which points toward a useful direction for future research. The sample period (2 January 2015 to 31 March 2026) covers a single phase of Bitcoin's history, the institutional-adoption decade. Bitcoin in earlier periods was thinly traded, dominated by retail flows, and behaved very differently. External validity beyond this sample is therefore limited; the findings are valid for the institutional-Bitcoin era they describe and should not be extrapolated to an earlier or later regime without additional empirical work. A natural extension is to re-run the analysis on a longer post-1990 sample once enough post-ETF data has accumulated. The 10% allocation cap used in the grid search reflects current institutional norms but is higher than the 3% to 5% consensus that prevailed at the start of the sample period. The cap was informed by knowledge of how Bitcoin performed across the full sample, which is a soft form of look-ahead. A genuinely real-time analysis would have used a tighter cap and reached more conservative conclusions about the headline allocation. A future replication could enforce a time-varying cap that reflects institutional norms as they evolved during the sample. Transaction costs are not modelled. Institutional estimates of 30 to 100 basis points per year for a 6% Bitcoin allocation would meaningfully erode the reported Sharpe uplift. Subsequent work should run the same analysis net of an explicit cost model calibrated to a representative institutional cost structure, and could also stress the result against rebalancing frequency choices. The walk-forward out-of-sample analysis uses six annual test windows, the most the sample length allows. Six is a small number for a binomial persistence test, and the corresponding confidence intervals are wide. Future work with a longer sample, or with shorter test windows that accept the noisier Sharpe estimates that come with them, would have more statistical power on the persistence question. Bitcoin is chosen as the cryptocurrency under study because of its dominant market capitalisation and institutional acceptance. This introduces a form of survivorship bias: in 2015 there were several hundred candidate cryptocurrencies of which Bitcoin is the only one to have reached its current scale. A genuinely ex ante analysis would have to confront the question of which crypto asset to pick at the start of the sample, and Bitcoin would not have been the obvious answer at the time. Re-running the analysis on a market-capitalisation-weighted basket of cryptocurrencies would be a useful robustness exercise. Finally, the dynamic correlation model uses the vanilla DCC(1,1) specification. The asymmetric ADCC extension of Cappiello, Engle and Sheppard (2006) would let correlations respond differently to positive and negative shocks, and is the most useful refinement for future work on the correlation dynamics themselves. So is the more sophisticated regime classifier that the unsatisfactory result of the Section 4.4 dynamic-allocation check would require: a DCC-GARCH conditional correlation forecast in place of the backward-looking rolling estimator, perhaps combined with a multi-factor regime score that includes the VIX and credit-spread indicators.

Chapter 6: Conclusion

6.1 Summary of Findings

The research question asked whether incremental Bitcoin allocation to a traditional 60/40 portfolio improves its risk-adjusted return, and whether that improvement is statistically significant under nonstationarity, regime shifts, and resampling. The three sub-questions and their answers run as follows.

RQ1 asked which allocation maximises the Sharpe ratio. A grid search across allocations from 0% to 10% produced a monotonically increasing Sharpe curve, with the maximum at the 10% institutional cap (annualised Sharpe of 0.698 against a benchmark of 0.449, a 55.5% relative uplift). The conservative headline allocation adopted for downstream tests is $x^* = 6\%$, which delivers a +0.178 absolute (39.7% relative) Sharpe uplift and is the largest weight at which both significance tests reject the null cleanly.

RQ2 asked whether that uplift is statistically significant. At $x^* = 6\%$, the moving block bootstrap returns a one-sided p of 0.047 and the Deflated Sharpe Ratio of Bailey and López de Prado (2014) returns 0.954, both passing the 5% threshold. At $x = 10\%$ the DSR is 0.972 (clearly significant) and the bootstrap p is 0.074 (borderline).

RQ3 asked whether the uplift persists under regime conditioning and walk-forward out-of-sample validation. The regime answer is yes with a strong qualifier: the uplift is concentrated in the 56% of the sample where the BTC-SPX correlation is below 0.25 (regime Δ Sharpe of +0.49), and disappears or reverses in the high-correlation regime that accounts for the remaining 14%. The walk-forward answer is negative: a static rule trained on five years and applied to the following year produces a positive Δ Sharpe in only three of six test years, and the stitched out-of-sample bootstrap CI includes zero.

The CAPM regression confirms that Bitcoin has the statistical signature of a diversifier at the asset level (alpha of +39.3% per year, significant at the 5% level; beta of 0.84; R-squared of 0.052). The honest single-sentence summary is that the diversification benefit is real but regime-dependent, and a static allocation rule does not capture it reliably.

6.2 Contributions to Knowledge

The contribution of this dissertation is more methodological than empirical. The empirical question (does small Bitcoin improve a 60/40 portfolio?) has been asked many times before, with broadly affirmative answers. What is new here is the combination of four inference procedures applied to the same dataset, the same allocation grid, and the same research question. None of the four procedures is novel on its own, but their joint application to the Bitcoin diversification question, in a way that lets each procedure check the others, has not been standard in the prior literature.

Three specific contributions follow. The dissertation applies the Deflated Sharpe Ratio of Bailey and López de Prado (2014) to a Bitcoin-augmented 60/40 portfolio, which has not been a standard inference choice in the Bitcoin diversification literature. It reports the Sharpe-uplift estimate conditional on three correlation regimes defined by both a rolling 60-day classifier and a DCC-GARCH parametric companion, rather than as a single full-sample average. And it subjects the in-sample finding to a six-window walk-forward out-of-sample test and reports the result transparently, including the finding that the static-allocation benefit does not transfer reliably.

The empirical contribution that follows is a more nuanced reading of the Bitcoin diversification question than the institutional headline figures suggest. The summary that emerges (a real but regime-dependent benefit that requires dynamic harvesting) is more useful to an institutional investment committee than the unconditional Sharpe uplift typically reported by the industry research papers, because it identifies the market conditions under which the historical uplift is likely to hold prospectively.

6.3 Directions for Future Research

The limitations in Section 5.4 each point to a useful direction for further work. Five are worth singling out.

The most pressing is the dynamic-allocation extension. The exploratory rule tested in Section 4.4 ($x = 10\%$ when the lag-1 rolling correlation is below 0.25, $x = 5\%$ in the moderate regime, $x = 0\%$ in the high regime) improved the stitched out-of-sample Sharpe meaningfully but did not pass the binomial persistence test cleanly. A more sophisticated rule built on a DCC-GARCH conditional correlation forecast, rather than on a backward-looking rolling estimator, would be the natural next step. A further refinement would condition on a multi-factor regime score that includes the VIX, credit spreads, and the term-structure slope alongside the BTC-SPX correlation, with the regime score updated in real time.

A second avenue is to replicate the analysis net of an explicit transaction-cost model. Institutional estimates of 30 to 100 basis points per year for a 6% Bitcoin allocation would meaningfully erode the reported Sharpe uplift, and the breakeven cost at which the in-sample benefit disappears is itself an interesting empirical question.

Extending the comparison set beyond Bitcoin is another natural step. The dissertation does not compare Bitcoin against gold, commodities, hedge funds, or other alternative diversifiers, and the practical-implications discussion would be considerably stronger if it did. A market-capitalisation-weighted basket of cryptocurrencies, replacing Bitcoin in the same framework, would also address the survivorship-bias concern raised in Section 5.4.

Widening the geographic scope would test whether the regime-conditional finding generalises beyond US assets. The analysis here uses the S&P 500, the US Aggregate Bond Index, and a US Treasury Bill rate, because Bitcoin pricing is dollar-denominated and the institutional Bitcoin literature has been US-led. A similar analysis applied to a European 60/40 portfolio, using a Euro-area equity index, a Euro-denominated bond index, and the three-month EURIBOR rate as the risk-free benchmark (the latter already present in the master dataset assembled for this dissertation), would test whether the regime-conditional finding holds in a different currency and market environment.

Finally, extending the sample as more post-ETF data accumulates would sharpen the inferential picture. The January 2024 spot Bitcoin ETF approval is one of the two structural breaks confirmed in this dissertation, and there is now only about two years of post-break data. Re-running the analysis on a longer post-ETF sample, ideally after another five years have elapsed, would let the inferential procedures used here speak more directly to the post-ETF Bitcoin market that institutional investors actually face today.

6.4 Concluding Remarks

The starting point of this dissertation was a simple practical question: does adding a small Bitcoin allocation to a traditional 60/40 portfolio improve risk-adjusted returns enough to justify the asset's place in an institutional mandate? The institutional literature has tended to give a confident yes. The academic literature has tended to give a qualified yes. The contribution here is to give a more careful yes-and-no answer that reflects the data more honestly.

Bitcoin is a genuine diversifier at the asset level. Its equity beta is below one, its idiosyncratic variation accounts for 95% of its return volatility, and its alpha against the S&P 500 is positive and

statistically significant. On any cross-sectional measure of diversification potential, the asset qualifies. The in-sample Sharpe-ratio uplift of adding it to a 60/40 portfolio is also real and statistically significant under the most rigorous of the inference tests applied here.

The qualification is that the benefit is regime-dependent and does not transfer reliably out-of-sample under a static allocation rule. The static-rule view that has dominated industry research is incomplete. The honest practical implication is that institutional investors who want to harvest the documented benefit need to allocate dynamically in response to the prevailing correlation regime, accept the operational complexity that comes with it, and budget for the transaction costs and behavioural-execution challenges of conditioning weights on a noisy real-time signal.

If the result of this dissertation is a more conservative and conditional answer than the one the institutional literature has been giving, that is the most useful thing a careful empirical analysis could have produced. The data does not support an unconditional positive recommendation. It does support a regime-conditional one, with appropriate caveats, and those caveats are what an investment committee actually needs to know.

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AI tools were used to assist with the paper's formatting and to generate preliminary statistical graphs, tables, and the flow chart. The author also received help in drafting a few sentences during the writing process to make the paper sound more academically focused. All outputs were reviewed, edited, and verified by the author.

Appendices

Appendix A: Master Dataset Construction Notes

The empirical analysis uses a master daily panel constructed from five publicly available data sources, aligned on the US equity trading calendar over the period 2 January 2015 to 31 March 2026. This appendix documents the data sources, alignment conventions, and treatment choices made in constructing the panel referenced in Chapter 3 and used throughout Chapters 4 and 5.

Sources S&P 500 daily close prices were obtained from publicly downloaded historical files reflecting the index publisher. The Bloomberg US Aggregate Bond Total Return Index (ticker LBUSTRUU) was obtained from a Bloomberg extract supplied with the dissertation dataset. Bitcoin daily close prices (BTC/USD) were obtained from CoinDesk historical files. The three-month US Treasury Bill secondary-market rate (FRED series DTB3) was obtained from the Federal Reserve Economic Database. The CBOE Volatility Index (VIX) was obtained from a Bloomberg extract.

Calendar alignment: All series were merged on the US equity trading calendar, which produced 2,827 trading days over the sample period. The first row was dropped after log returns were computed, because the first daily return is undefined; this leaves 2,826 daily observations for inferential analysis.

Forward-fill treatment. The US Aggregate Bond Index and the three-month T-Bill rate are not published on certain US federal holidays. Where the underlying source did not publish a value on a US equity trading day, the previous published value was forward-filled. The number of forward-filled observations is small (fewer than 130 days for the T-Bill rate over the full sample, and roughly 36 for the bond index).

Bitcoin weekend handling: Bitcoin trades on a 24/7 calendar. Weekend prices were excluded from the master panel to align with the US equity trading calendar convention used in the prior literature (Brière et al., 2015). This is the standard convention and was preferred over interpolation because it preserves the actual daily returns experienced by an institutional investor whose portfolio is rebalanced on US trading days.

Outlier treatment: No winsorisation or other outlier treatment was applied. The two most consequential extreme observations are the COVID-19 shock of March 2020 and the May 2022 cryptocurrency selloff, both of which are central to the regime analysis in Chapter 4. Removing them would distort the regime-conditional findings that are the heart of the dissertation.

Return computation: All returns are daily log returns, computed as $\ln(P_t / P_{t-1})$. Log returns are used for additive aggregation across time and for distributional symmetry. Annualised return is the daily mean multiplied by 252. Annualised volatility is the daily standard deviation multiplied by the square root of 252. Annualised Sharpe ratio is (annualised return minus annualised risk-free rate) divided by annualised volatility, with the risk-free rate proxied by the DTB3 series converted from annualised percent to daily decimal.

Appendix B: Full Grid-Search and Significance Results

Table 3 in Chapter 4 reports the headline significance results at the two reportable allocations ($x^* = 6\%$ and $x = 10\%$). This appendix presents the full grid search across all eleven candidate allocations from 0% to 10% Bitcoin in 1% increments, together with the corresponding annualised Sharpe ratios, Δ Sharpe versus the 60/40 benchmark, block bootstrap one-sided p-values, two-sided 95% confidence intervals, and Deflated Sharpe Ratios. The detailed table appears below as Table B1.

BTC weight	Sharpe	Δ Sharpe	Bootstrap p (1-sided)	Bootstrap 95% CI	DSR
0% (benchmark)	0.449	—	—	—	—
1%	0.484	+0.035	0.020	[+0.001, +0.070]	0.889
2%	0.517	+0.068	0.025	[+0.000, +0.139]	0.908
3%	0.548	+0.099	0.032	[−0.002, +0.204]	0.924
4%	0.577	+0.128	0.035	[−0.006, +0.267]	0.936
5%	0.603	+0.155	0.041	[−0.011, +0.327]	0.946
6% (x^* , headline)	0.627	+0.178	0.047	[−0.018, +0.383]	0.954
7%	0.649	+0.200	0.051	[−0.029, +0.436]	0.960
8%	0.667	+0.219	0.059	[−0.039, +0.489]	0.965
9%	0.684	+0.235	0.067	[−0.048, +0.528]	0.969
10% (Sharpe-max)	0.698	+0.249	0.074	[−0.060, +0.568]	0.972

Table B1. Full grid search across allocations 0% to 10% Bitcoin. Bootstrap parameters: $n = 1,000$, block = 5 days, seed = 42. Deflated Sharpe Ratio (Bailey and López de Prado, 2014) corrects for skewness, kurtosis, sample length, and the multiple-testing bias across $K = 10$ candidate trials. Source: author.

Two patterns visible in Table B1 are worth flagging. First, the Deflated Sharpe Ratio crosses the conventional 0.95 rejection threshold first at $x = 6\%$ and remains above it through $x = 10\%$. Second, the block bootstrap one-sided p-value remains below 0.05 from $x = 1\%$ through $x = 6\%$, but crosses the threshold between $x = 6\%$ and $x = 7\%$. The intersection of both criteria, that is, the largest allocation at which both significance procedures pass the 5% level, is $x = 6\%$, which is the headline allocation adopted in the main text.

The widening of the bootstrap confidence intervals at higher allocations reflects the well-known scaling of the Sharpe estimator's sampling variance with portfolio volatility: as the Bitcoin weight rises, portfolio volatility rises, and the same number of bootstrap replications produces a wider sampling distribution. The Deflated Sharpe Ratio uses an analytic correction that does not exhibit this scaling, which is why the two procedures diverge at the higher allocations.